

SQL DATABASE REPORT

WEEK – 3 TASK

Plant Management System

SQL Queries, Reports and Output

Scenario

The Plant Management System stores information about plants, their categories, prices and available stock quantities. It also maintains customer orders and payment information. SQL queries are used to display plant details, count plants by category, analyse prices, combine customer and order information, and display payment details.

Objectives

Display plant information and stock quantity.

Find the number of plants in each category.

Find the highest and lowest plant prices.

Combine Customer and Orders information using a JOIN.

Display order payment method and payment status.

Database Information Used in the Reports

Plant: Plant_ID, Plant_Name, Category, Price, Stock_Quantity

Customer: Customer information including Full_Name and Customer_ID

Orders: Order_ID, Order_Date, Total_Amount, Order_Status and Customer_ID

Payment: Order_ID, Payment_Method and Payment_Status

1. Plant Details Report

Purpose

This report displays the plant ID, plant name, category, price and available stock quantity.

```
SQL> SELECT Plant_ID, Plant_Name, Category, Price, Stock_Quantity
2 FROM Plant;
```

PLANT_ID	PLANT_NAME	CATEGORY	PRICE	STOCK_QUANTITY
101	Rose Plant	Flowering	250	20
102	Aloe Vera	Medicinal	180	15
103	Money Plant	Indoor	150	30
104	Tulsi Plant	Medicinal	120	25

Output: Plant Details

The output displays four plant records: Rose Plant, Aloe Vera, Money Plant and Tulsi Plant, along with their categories, prices and stock quantities.

2. Category-Wise Plant Count Report

Purpose

This report displays the number of plants available in each category by grouping the records according to Category.

```
SQL> SELECT Category, COUNT(*) AS Number_Of_Plants
2 FROM Plant
3 GROUP BY Category;
```

CATEGORY	NUMBER_OF_PLANTS
Indoor	1
Medicinal	2
Flowering	1

SQL> |

Output: Category-Wise Plant Count

The output shows: Indoor – 1 plant, Medicinal – 2 plants, and Flowering – 1 plant.

3. Plant Price Analysis

Purpose

This report identifies the highest and lowest prices recorded in the Plant table.

```
SQL> SELECT MAX(Price) AS Highest_Price,
2          MIN(Price) AS Lowest_Price
3          FROM Plant;

HIGHEST_PRICE  LOWEST_PRICE
-----
250            120

SQL> |
```

Output: Highest and Lowest Plant Price

The output shows a highest price of 250 and a lowest price of 120.

4. Customer Order Report

Purpose

This report combines Customer and Orders information. It displays the customer's full name together with order ID, order date, total amount and order status.

```
SQL> SELECT c.Full_Name,
2          o.Order_ID,
3          o.Order_Date,
4          o.Total_Amount,
5          o.Order_Status
6          FROM Customer c
7          JOIN Orders o
8          ON c.Customer_ID = o.Customer_ID;
```

FULL_NAME

ORDER_ID ORDER_DAT TOTAL_AMOUNT

ORDER_STATUS

Ravi Kumar
1001 20-SEP-26 430
Confirmed

Priya Sharma
1002 21-SEP-26 150
Delivered

FULL_NAME

ORDER_ID ORDER_DAT TOTAL_AMOUNT

ORDER_STATUS

Ravi Kumar
1003 22-SEP-26 250
Pending

Output: Customer and Order Details

The output contains orders for Ravi Kumar and Priya Sharma. It shows the corresponding order dates, amounts and statuses such as Confirmed, Delivered and Pending.

5. Payment Report

Purpose

This report displays the payment method and payment status associated with each order.

```
SQL> SELECT Order_ID, Payment_Method, Payment_Status
2 FROM Payment;

ORDER_ID PAYMENT_METHOD
-----
PAYMENT_STATUS
-----
1001 UPI
Paid
1002 Cash on Delivery
Paid
1003 Credit Card
Pending
```

Output: Payment Details

The output shows UPI, Cash on Delivery and Credit Card as payment methods. The corresponding payment statuses are Paid, Paid and Pending.

Conclusion

The SQL reports demonstrate how a plant management database can be used to retrieve and analyse information. SELECT queries display records, GROUP BY with COUNT() provides category-wise totals, MAX() and MIN() analyse prices, JOIN combines related customer and order information, and the Payment query displays payment details.

Summary of Reports

Report	SQL Concept	Purpose
Plant Details	SELECT	Display plant and stock information
Category-Wise Count	GROUP BY, COUNT()	Count plants in each category

Price Analysis	MAX(), MIN()	Find highest and lowest prices
Customer Order	JOIN	Combine customer and order information
Payment	SELECT	Display payment method and status